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PUBLICATION OF THE PATENT OFFICE

INTRODUCTION

In view of the recent amendment made in the Patents Act, 1970 by the Patents (Amendment) Act, 2005 effective from 01st January 2005, the Official Journal of The Patent Office is required to be published under the Statute. This Journal is being published on weekly basis on every Friday covering the various proceedings on Patents as required according to the provision of Section 145 of the Patents Act 1970. All the enquiries on this Official Journal and other information as required by the public should be addressed to the Controller General of Patents, Designs & Trade Marks. Suggestions and comments are requested from all quarters so that the content can be enriched.

(PROF. (DR) UNNAT P. PANDIT)
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22th May, 2026

CONTENTS

<i>SUBJECT</i>		<i>PAGE NUMBER</i>
JURISDICTION	:	58972-58973
SPECIAL NOTICE	:	58974-58975
EARLY PUBLICATION (DELHI)	:	58976-59379
EARLY PUBLICATION (MUMBAI)	:	59380-59720
EARLY PUBLICATION (CHENNAI)	:	59721-60561
EARLY PUBLICATION (KOLKATA)	:	60562-60658
PUBLICATION AFTER 18 MONTHS (DELHI)	:	60659-61131
PUBLICATION AFTER 18 MONTHS (MUMBAI)	:	61132-61166
PUBLICATION AFTER 18 MONTHS (CHENNAI)	:	61167-61407
PUBLICATION AFTER 18 MONTHS (KOLKATA)	:	61408-61448
WEEKLY ISSUED FER (DELHI)	:	61449-61601
WEEKLY ISSUED FER (MUMBAI)	:	61602-61650
WEEKLY ISSUED FER (CHENNAI)	:	61651-61758
WEEKLY ISSUED FER (KOLKATA)	:	61759-61776
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (DELHI)	:	61777-61800
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (MUMBAI)	:	61801-61812
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (CHENNAI)	:	61813-61829
PUBLICATION UNDER SECTION 43(2) IN RESPECT OF THE GRANT (KOLKATA)	:	61830-61835
PUBLICATION U/R 84(3) IN RESPECT OF APPLICATION FOR RESTORATION OF PATENTS	:	61836
INTRODUCTION TO DESIGN PUBLICATION	:	61837
DESIGN CORRIGENDUM	:	61838
REGISTRATION OF DESIGNS	:	61839-62155

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(54) Title of the invention : COAL MINE HELMET LIVE TRACKING AND SAFETY SYSTEM BASED ON IOT WITH UWB LOCALIZATION, LORA RELAY COMMUNICATION, AND MULTI-SENSOR HAZARD DETECTION

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(57) Abstract :

Coal mining has remained one of the most hazardous industries due to risks including toxic gas leakage, high temperature, fire accidents, poor communication, and difficulty in locating miners inside underground tunnels. This invention provides a Coal Mine Helmet Live Tracking and Safety System based on IoT technologies. The smart helmet includes sensors for gas detection, temperature monitoring, flame detection, and an emergency push button. A UWB (Ultra-Wideband) tag mounted on the helmet enables accurate real-time miner location tracking inside tunnels through UWB anchor nodes installed underground. The collected sensor data and location information are processed using a microcontroller such as ESP32 or STM32. For long-distance communication from tunnel to surface, LoRa wireless modules transmit data through relay nodes and a LoRa gateway. The gateway sends the data to a cloud server or control room dashboard via Wi-Fi, Ethernet, or 4G/LTE network. The monitoring system displays miner positions on a live map and continuously checks environmental conditions. When gas leakage, fire, abnormal temperature, or emergency button activation occurs, instant alerts generate through alarms, SMS, voice calls, email, and mobile application notifications. This system provides reliable underground communication, accurate live tracking, and early hazard detection, reducing accident risks and enabling faster rescue operations. The system presents a cost-effective and scalable solution for modern coal mine worker safety management.

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